Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/15/2010
PI Name:	Cogan, Stuart Sc.D.		
Project Title:	Flexible High-Barrier Polymers for Food Packaging		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02062-2612	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	12/24/2009	**End Date:**	05/16/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Sullivan, Thomas	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/16/2012 (previously 12/23/2011) per 3/14/2012 HRP Master Task List information (Ed., 4/27/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10CB04C		
Performance Goal No.:			
Performance Goal Text:			
	The development of a polymer laminate with water and oxygen barrier properties suitable for food packaging and preservation on 3-5 year manned space exploration missions is proposed. The laminate is a multilayer structure comprising polymer and inorganic dielectrics that will provide near-hermetic encapsulation of food items for the duration of these missions. In Phase I, flexible polymer barriers with an oxygen transport rate of <0.005 cc/m2-day and water transport rate of <0.005 g/m2-day were developed. The barriers contain no metal foils, have a areal density of <34 g/m2 for a 40 micron thick film, and tolerate high temperature sterilization treatments. The polymer laminates are mechanically robust exhibiting a 165MPa yield strength, 200MPa tensile strength, 550MPa tensile modulus, and 3% elongation to yield. In Phase II, we propose to optimize barrier properties to reduce weight, minimize ash on incineration, develop heat-sealing methods, and expand the testing to include heat sealed enclosures. The Phase II effort also includes		

Task Description:	a collaboration with a potential high-volume manufacturer of the barrier films for aerospace applications. POTENTIAL NASA COMMERCIAL APPLICATIONS: The specific NASA applications for the proposed barrier polymers are the following: * food packaging for long-duration space missions; * packaging for medicines on long-duration space missions; * packaging of air and moisture sensitive chemicals used in science exploration on long-duration missions; *packaging of astronaut physiological samples; *packaging of extraterrestrial materials for return to earth. All these applications will benefit from the availability of a lightweight, high-strength flexible packaging material. Besides having exceptional barrier properties, the proposed barrier polymers have several other features that are important to these NASA applications including: 1) transparency to allow inspection and identification of package contents; 2) an internal surface to which biocompatible and non-thrombogenic coatings can be added; and 3) a low temperature heat sealing capability that allows astronauts in space to create sealed packages that exclude water and oxygen and also prevent bacteriological and viral contamination.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Non-Nasa commercial applications in both the private and Government sectors have been identified. These applications included: * meals-ready-to-eat (MREs) packaging for the military *electronics packaging * flexible packaging for primary and secondary lithium, ion batteries * packaging of battlefield medical supplies for treating trauma (e.g. biologically active hemostatic agents) * very long storage of food and medicines in rapid response materiel caches * packaging emergency supplies at remote locations.
Task Progress:	New project for FY2010. Reporting not required for this SBIR Phase 2 project. (Note: task added to Task Book in November 2010 when received information.)
Bibliography Type:	Description: (Last Updated:)